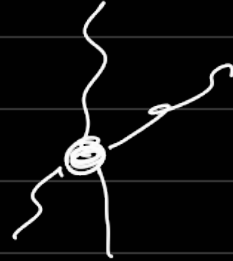
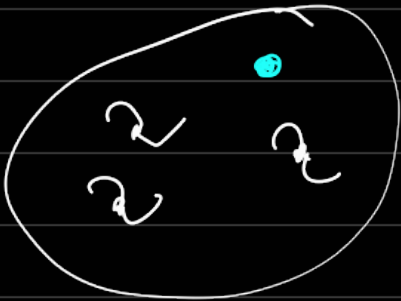


* Bar pressure & ICM/IGM:

- Similar to gal: total mass, total lum.
- $\left(\frac{M}{L}\right)$: tell us about P.M.

- Assume: Hydrogen inside Gal cluster



$$m_p: U_{\text{clust.}} = - \frac{G M_{\text{clust}} m_p}{R} \quad *$$

- Sim. ell. we assume vir eq: $2K + U = 0$

$$K = -\frac{1}{2}U$$

$$m_p: \frac{M}{2} K T \rightarrow \frac{3}{2} K T \stackrel{\text{clust.}}{=} K$$

$$\frac{3}{2} K T = -\frac{1}{2} \left(- \frac{G M_{\text{clust}} m_p}{R} \right) \quad *$$

$$* (T_{\text{clust}}) = \frac{G M_{\text{clust}} m_p}{3 R K} \quad *$$

→ Electrons do not exper. grav. eq.

- Arg. protons will interact w/ e^- and give half K.E.

⇒ Temp is Half



$$(T_{\text{clust}})_{\text{req.}} = \frac{1}{2} (T_{\text{clust}}) = \frac{G M_{\text{clust}} m_p}{6 R K} \quad *$$

$$M_{\text{clust}} = \frac{(T_{\text{clust}})_{\text{req.}} 6 R K}{G m_p} \quad (\text{Method 1})$$

(Method 2): (Mass)

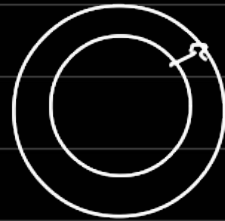
$$\tilde{r}_e^2 = 3 \sigma_e^2$$

• Obs. vel. d.f. | $\sigma_{\text{obs}}^2 = \langle V_{\text{gal}}^2 - \langle V_{\text{gal}} \rangle^2 \rangle$

• Vir thm, $2K + U = 0$

$$K = \frac{1}{2} M_{\text{gal}} (\tilde{\sigma}^2) = \frac{3}{2} M_{\text{gal}} \sigma^2 = -\frac{1}{2} U_{\text{gal}}$$

$$U_{\text{gal}} = -\alpha \frac{G M_{\text{gal}}^2}{R} \leftarrow$$



$$\frac{3}{2} M_{\text{gal}} \sigma^2 = \frac{\alpha}{2} \left(\frac{G M_{\text{gal}}^2}{R} \right)$$

solve M_{gal}

$$M_{\text{gal}} = \frac{3}{\alpha} \frac{\sigma^2 R}{G}$$

* (vel. of gal)

* We want L_{vir} "given":

• Eff. luminosity

$$\epsilon_{\text{ff}} = 1.48 \times 10^{-40} n_e^2 T^{4/2} (1.2) \text{ W m}^{-3}$$

$$[\epsilon_{\text{ff}}] = \frac{L}{(\text{Vol})} \quad | \quad \text{Vol} = \frac{4\pi}{3} R^3$$

$$L_X = R^3 7.44 \times 10^{-40} n_e^2 T^{1/2}$$

* n_e, T

* ↑ are above

$$\left(\frac{M}{L} \right)$$

• Typically: $M_{\text{clust}} \approx 10^{15} M_{\odot}$ *

$$L \approx 10^{13} M_{\odot}$$

$$\left(\frac{M}{L} \right) \sim 10^2 \sim 100 \left(\frac{M_{\odot}}{L_{\odot}} \right) \leftarrow 50 \sim 500 \left(\frac{M_{\odot}}{L_{\odot}} \right)$$

* $\left(\frac{M}{L} \right) \sim 2 \left(\frac{M_{\odot}}{L_{\odot}} \right)$ (Stellar - M-to-L ratio)

• near the sun

$$\star \left(\frac{M}{L} \right) \sim 20-50 + \left(\frac{M_{\odot}}{L_{\odot}} \right) (\text{Rad.} + \text{D.M.})$$

• 20 +

$\sim 7\%$ gal. $\star$

$\star$ D.M. accounts for $\sim 95\%$ $\leftarrow$
Rad. matter $\sim 5\%$

Might ask: IGM add enough of mass

$\rightarrow$ It does not

$$M(\text{IGM}) \simeq (0.1-0.2) M_{\text{cluster}} \simeq \underline{10^{14} M_{\odot}}$$

• ISM: at Schrö. radius

$$M_{\text{ISM}} \simeq \underline{\underline{10 M_{\odot}}} \text{ at } r_{\text{Sch}}.$$

PS6

① $R = 1.5 \text{ Mpc}$ •

$T = 7 \times 10^7 \text{ K}$ •

② $L = 1.5 \times 10^{36} \text{ W}$ •

(i)
$$M_{\text{evst}} = \frac{T G R n}{G_{\text{mp}}}$$

$\star$

(ii) Also: use L :

$$L_{\text{K}} = R^3 7.44 \times 10^{40} n e^2 T^{1/2}$$

solve for n :

$$\rho = m n e = \frac{M_{\text{tot}}}{\text{Vol}}, \quad n_e = \frac{\#}{\text{Vol}}$$

• $M_{\text{tot}} = m_p n_e \text{Vol} = m_p n e \frac{4\pi}{3} R^3$ $\leftarrow$ $\star$

③ $1 - 1.2 \times 10^{12}$

⑤ $L_V = 1.2 \times 10^{40}$ W

* From our lec. $\left[\frac{M}{L}\right] \approx 50 - 500$

• assume some reasonable $\left(\frac{M}{L}\right)$ and then

$$\left(\frac{M}{L}\right) \approx 50 - 500$$

these are extreme ends

generally $\left(\frac{M}{L}\right) \approx 10 - 30$ "reasonable"

$$\text{assume } \left(\frac{M}{L}\right) \approx 10 - 30 \quad \left| \quad \frac{1}{T} = \left(\frac{\partial S}{\partial E}\right) \right.$$

$$(M_{\text{tot}})_{\text{DM}} = (10 - 30) L_V \quad \left. \begin{array}{l} S = k_B \ln(\Omega) \end{array} \right\}$$

ex: mp: $\frac{3}{2} kT = \frac{1}{2} m \underbrace{V_{\text{rms}}^2}$

⑥ $\sigma_r = 666 \text{ km/s}$ *

$$M_{\text{cusp}} = \frac{3}{a} \frac{\sigma_r^2 R}{G} = \frac{3 \sigma_r^2 R}{G}$$

* Just assume $a=1$

Plug in.

⑦ $t_{\text{cross}} = \frac{R}{(V_{\text{el}})} = \frac{R}{\sqrt{\sigma_r^2}} = 1.3 \times 10^9 \text{ yr}$

$V_{\text{el}} \equiv \sqrt{\sigma_r^2}$

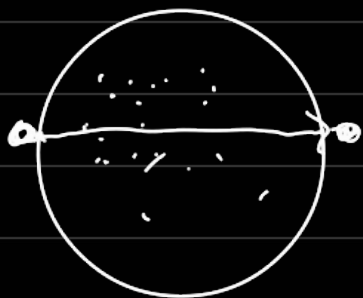
$$V_{\text{obs}} = \underbrace{V_{\text{Hubble}}}_{V_{\text{Hubble}}} + V_{\text{pec}}$$

$$V_{\text{Hubble}} = H_0 D$$

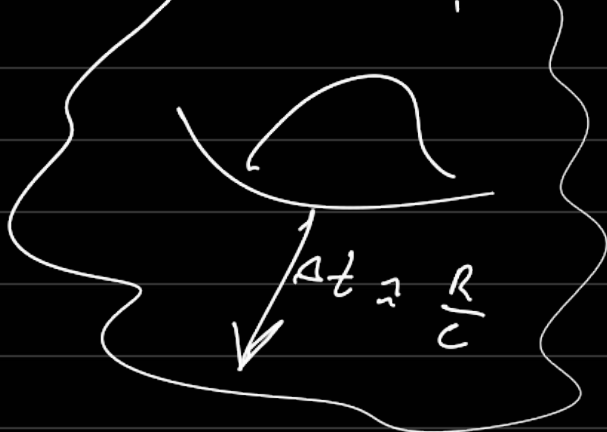
↑ Hubble time $\sim \frac{1}{H_0}$

$H_0^{-1} \approx$ Age of the universe today

$$\approx 13.7 \times 10^9 \text{ yrs}$$



BH



(2) (a) $2u + v > 0$

$$U = - \frac{G M_{\text{Holo}} m_p}{R}$$

$$T_{\text{Holo}} = \frac{G M_{\text{Holo}} m_p}{6 R k}$$



* Similar to B.H. sphere
 $\Rightarrow$ Kepler orbit

$$v^2 = \frac{GM}{R}$$

$$T_{\text{Holo}} = \frac{v^2 m_p}{6 k}$$

assumes most of the "constituents" are on avg m_p

* Get c temp $\rightarrow$ temp to

$$E_\gamma = h\nu \approx kT \Rightarrow \boxed{T = \frac{h\nu}{k}}$$

$$T \approx 1.3 \times 10^6 \text{ K}$$

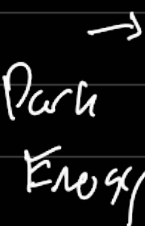
find

$$c = v\lambda$$

v or λ

Short Section:

* Look at large scale $\rightarrow$ small scale



* Found How D.M. Demolishes and creates the structure we see today

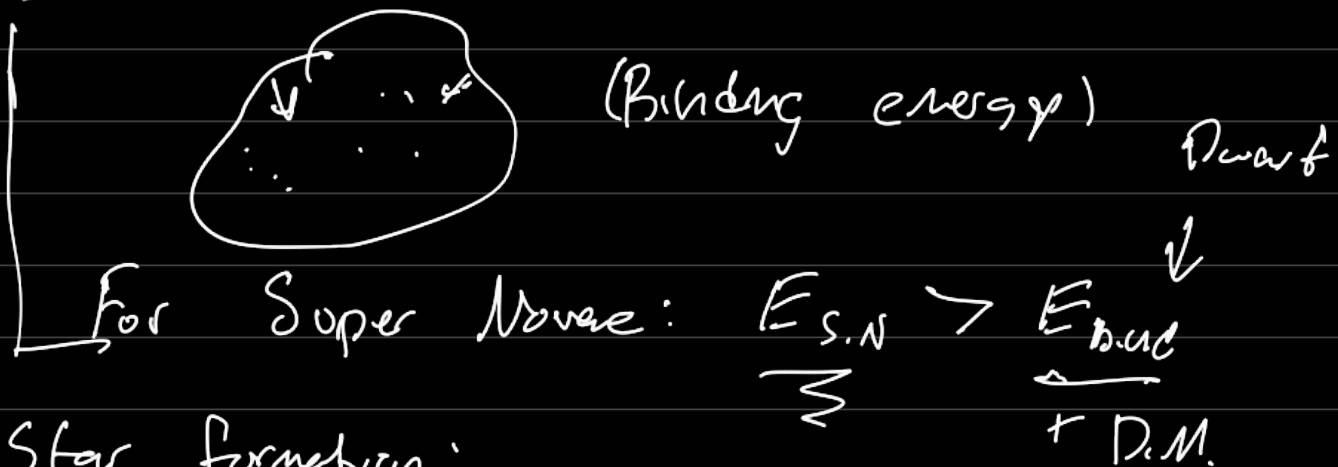
CMB: $z \approx 1100$

- Tells us the Initial conditions to how the universe today

* D.M: How galaxies form $\Rightarrow$ Star formation

- D.M + Mechanics explain what we or lack here of

ex: But see Mary Dewarfigal.



Star formation:



• Radiation: rad. pressure

• Form $F_{\text{grav}} > P_{\text{rad}}$ ✗

ex: Using free-fall times



$$\frac{P^2}{a^3} = \frac{4\pi^2}{GM}$$

$$R = 2a$$

$$t_{\text{ff}} = \frac{P}{2}$$

Key:

$$\rho_{\text{gas}} = \frac{M}{\text{vol}}$$

$$\text{vol} = \frac{4\pi}{3} R^3$$

$$M = \frac{4\pi}{3} R^3 \rho_{\text{gas}}$$

Plug in

$$(t_{\text{ff}}) = \left(\frac{3\pi}{32\rho_{\text{gas}} G} \right)^{1/2}$$

